

Supplementary Material

S1: Statistical Analysis and Reporting of Experimental Results

Table 1: Experimental performance comparison of different models on various tasks in ADNI, where " " denotes the second-best model. Notably, (M) denotes a multimodal method, and # indicates a statistically significant result.

Tasks	AD vs NC (ADNI)				AD vs MCI (ADNI)				NC vs MCI (ADNI)			
Model	ACC	F1	REC	PRE	ACC	F1	REC	PRE	ACC	F1	REC	PRE
sMRI												
3D ResNet	0.885±0.015 #	0.885±0.014	0.872±0.017	0.897±0.013	0.803±0.021 #	0.803±0.019	0.812±0.023	0.791±0.019	0.681±0.019 #	0.682±0.024	0.690±0.025	0.681±0.022
DA-MIDL	0.862±0.017 #	0.861±0.021	0.852±0.020	0.871±0.017	0.788±0.027 #	0.787±0.025	0.776±0.021	0.797±0.027	0.659±0.027 #	0.662±0.021	0.672±0.029	0.657±0.019
Gao et al.	0.870±0.023 #	0.870±0.017	0.881±0.017	0.863±0.019	0.793±0.021 #	0.793±0.021	0.793±0.023	0.793±0.020	0.669±0.025 #	0.658±0.031	0.648±0.023	0.680±0.034
Segmamba*	0.892±0.021	0.891±0.014	0.878±0.022	<u>0.903</u> ±0.021	0.815±0.023 #	0.815±0.024	0.820±0.027	0.811±0.023	0.706±0.020 #	0.699±0.027	0.682±0.027	0.718±0.025
UNETR*	0.883±0.020 #	0.883±0.019	0.881±0.022	0.885±0.019	0.801±0.025 #	0.801±0.023	0.801±0.023	0.801±0.023	0.702±0.031 #	0.695±0.033	0.678±0.036	0.712±0.031
VT-UNet*	0.888±0.021 #	0.888±0.021	0.892±0.018	0.883±0.023	0.811±0.009 #	0.811±0.016	0.817±0.011	0.805±0.019	0.705±0.023 #	0.703±0.029	0.698±0.027	0.710±0.030
TransBTS*	0.894±0.015 #	0.894±0.018	0.897±0.017	0.889±0.019	0.819±0.019 #	0.818±0.020	0.815±0.019	0.821±0.022	0.722±0.024 #	0.714±0.021	0.697±0.021	0.737±0.027
MCADS	0.898±0.023	0.896±0.021	0.900±0.025	0.8929±0.013	0.838±0.024 #	0.835±0.027	<u>0.844</u> ±0.033	0.826±0.027	0.709±0.029 #	0.703±0.027	0.698±0.028	0.708±0.026
GF-Net	0.888±0.019 #	0.886±0.017	0.881±0.021	0.891±0.014	0.811±0.022 #	0.806±0.017	0.810±0.018	0.802±0.017	0.706±0.033 #	0.703±0.031	0.709±0.035	0.697±0.030
MAD-Former	<u>0.900</u> ±0.021	0.897±0.019	0.895±0.018	0.890±0.019	0.840±0.019 #	0.837±0.020	0.841±0.019	0.832±0.022	0.724±0.027 #	0.721±0.029	0.725±0.031	0.716±0.030

AAGN	0.882±0.019#	0.881±0.016	0.887±0.017	0.874±0.015	0.819±0.020 #	0.821±0.019	0.825±0.023	0.817±0.017	0.718±0.021 #	0.722±0.020	0.727±0.024	0.717±0.020
dGLCN	0.870±0.011#	0.867±0.012	0.872±0.012	0.861±0.011	0.823±0.017 #	0.823±0.019	0.823±0.022	0.823±0.020	0.712±0.035 #	0.715±0.037	0.720±0.031	0.710±0.035
IMDL(M)	0.887±0.027	0.886±0.022	0.876±0.024	0.896±0.021	<u>0.853</u> ±0.019	<u>0.853</u> ±0.023	0.853 ±0.017	<u>0.853</u> ±0.026	<u>0.755</u> ±0.031	<u>0.751</u> ±0.029	<u>0.739</u> ±0.033	<u>0.766</u> ±0.027
NeurGNN (ours)	0.912 ±0.002	0.912 ±0.002	0.912 ±0.024	0.912 ±0.024	0.869 ±0.009	0.865 ±0.008	0.841±0.020	0.891 ±0.029	0.756 ±0.024	0.753 ±0.014	0.740 ±0.017	0.767 ±0.049
NeurGNN (CI)	[0.9086, 0.9154]	[0.9094, 0.9146]	[0.8819, 0.9421]	[0.8824, 0.9424]	[0.8573, 0.8807]	[0.8559, 0.8757]	[0.8167, 0.8665]	[0.8551, 0.9277]	[0.7256, 0.7868]	[0.7347, 0.7713]	[0.7195, 0.7617]	[0.7060, 0.8296]
PET												
3D ResNet	0.890±0.037 #	0.888±0.031	0.875±0.033	0.903±0.029	0.841±0.032 #	0.840±0.033	0.826±0.031	<u>0.859</u> ±0.033	0.666±0.041 #	0.699±0.040	<u>0.776</u> ±0.040	0.650±0.040
PCC-GAN	0.911±0.042	0.908±0.032	0.880±0.037	0.940 ±0.029	0.840±0.027 #	0.838±0.028	0.836±0.026	0.844±0.029	0.684±0.033 #	0.678±0.037	0.663±0.039	0.708±0.035
M3T	0.888±0.050	0.893±0.041	<u>0.932</u> ±0.036	0.860±0.047	0.838±0.036 #	0.845±0.025	0.880 ±0.027	0.813±0.023	0.728±0.037	0.723±0.034	0.725±0.037	0.740±0.033
AmyloidPETNet	0.859±0.041 #	0.859±0.039	0.859±0.043	0.862±0.038	0.816±0.039 #	0.813±0.037	0.802±0.040	0.831±0.037	0.611±0.041 #	0.643±0.040	0.765±0.039	0.600±0.040
SegMamba*	0.903±0.048	0.901±0.040	0.890±0.040	0.914±0.039	0.797±0.048 #	0.787±0.041	0.755±0.040	0.829±0.042	0.670±0.042 #	0.706±0.035	0.797 ±0.033	0.643±0.037
VT-UNet*	<u>0.919</u> ±0.027	0.919 ±0.031	0.922±0.029	<u>0.921</u> ±0.033	0.852±0.017 #	<u>0.846</u> ±0.015	0.819±0.020	0.878 ±0.015	<u>0.751</u> ±0.038	<u>0.743</u> ±0.037	0.761±0.026	<u>0.765</u> ±0.035
TransBTS*	0.899±0.024 #	0.898±0.027	0.893±0.027	0.907±0.026	0.836±0.018 #	0.837±0.019	0.845±0.019	0.834±0.019	0.699±0.044 #	0.706±0.044	0.733±0.049	0.709±0.039
MCADS	0.916±0.035	0.909±0.037	0.902±0.037	0.895±0.036	0.842±0.039 #	0.839±0.037	0.834±0.037	0.844±0.037	0.731±0.042	0.688±0.040	0.702±0.041	0.674±0.040
GF-Net	0.894±0.025 #	0.891±0.029	0.887±0.021	0.894±0.031	0.842±0.041 #	0.841±0.039	0.843±0.037	0.839±0.040	0.718±0.033 #	0.715±0.034	0.720±0.027	0.709±0.037
MAD-Former	0.916±0.027	<u>0.917</u> ±0.025	0.921±0.024	0.913±0.025	0.848±0.037 #	<u>0.846</u> ±0.035	0.846±0.038	0.846±0.032	0.731±0.031	0.733±0.031	0.737±0.028	0.729±0.029
AAGN	0.888±0.031 #	0.887±0.027	0.885±0.028	0.889±0.027	0.838±0.031 #	0.839±0.033	0.841±0.027	0.837±0.035	0.726±0.039	0.727±0.036	0.723±0.035	0.717±0.036
dGLCN	0.906±0.033	0.906±0.031	0.906±0.032	0.906±0.028	0.841±0.034 #	0.840±0.034	0.842±0.031	0.837±0.034	0.718±0.051	0.717±0.049	0.715±0.047	0.719±0.044
IMDL(M)	0.887±0.037 #	0.886±0.035	0.876±0.033	0.896±0.036	0.853±0.028 #	0.853 ±0.031	<u>0.853</u> ±0.027	0.853±0.030	0.755 ±0.037	0.751 ±0.036	0.739±0.035	0.766 ±0.036

NeurGNN (ours)	0.936 ±0.008	0.916±0.017	0.935 ±0.031	0.899±0.018	0.866 ±0.026	0.839±0.033	0.822±0.038	0.858±0.048	0.738±0.017	0.664±0.014	0.644±0.014	0.690±0.017
NeurGNN (CI)	[0.9249, 0.9471]	[0.8957, 0.9379]	[0.8971, 0.9745]	[0.8763, 0.9221]	[0.8341, 0.8995]	[0.7978, 0.8802]	[0.7741, 0.8703]	[0.7979, 0.9181]	[0.7160, 0.7600]	[0.6460, 0.6820]	[0.6263, 0.6617]	[0.6687, 0.7113]

Table 2: Experimental performance comparison of different models on various tasks in OASIS, where " _ " denotes the second-best model.

Tasks	AD vs NC (OASIS)			
Model	ACC	F1	REC	PRE
SMRI				
3D ResNet	0.794±0.035 #	0.809±0.051	<u>0.873</u> ±0.049	0.823±0.053
PCC-GAN	0.750±0.057 #	0.739±0.053	0.708±0.056	0.772±0.051
M3T	0.760±0.047 #	0.752±0.044	0.729±0.041	0.777±0.045
AmyloidPETNet	0.800±0.061	0.804±0.063	0.830±0.060	<u>0.850</u> ±0.065
SegMamba*	0.791±0.057	0.795±0.051	0.812±0.056	0.780±0.050
VT-UNet*	0.798±0.043	0.798±0.041	0.808±0.041	0.810±0.042
TransBTS*	<u>0.803</u> ±0.041	<u>0.813</u> ±0.039	0.858±0.046	0.774±0.035
MCADS	0.798±0.051	0.796±0.051	0.802±0.050	0.787±0.050
GF-Net	0.785±0.034 #	0.787±0.031	0.800±0.029	0.772±0.030
MAD-Former	0.800±0.037 #	0.805±0.034	0.825±0.033	0.782±0.038
AAGN	0.784±0.045 #	0.787±0.047	0.812±0.042	0.762±0.044
dGLCN	0.798±0.048	0.791±0.049	0.808±0.047	0.773±0.051

IMDL(M)	0.802±0.053	0.781±0.051	0.708±0.049	0.871 ±0.049
NeurGNN (ours)	0.809 ±0.031	0.814 ±0.039	0.889 ±0.037	0.838±0.033
NeurGNN (CI)	[0.7705, 0.8475]	[0.7970, 0.8790]	[0.8431, 0.9349]	[0.7656, 0.8624]
PET				
3D ResNet	0.828±0.041 #	0.825±0.047	0.809±0.045	0.847±0.047
PCC-GAN	0.801±0.034 #	0.792±0.040	0.777±0.035	0.821±0.042
M3T	0.845±0.054	0.851±0.047	0.877±0.045	0.833±0.045
AmyloidPETNet	0.716±0.088 #	0.757±0.092	<u>0.878</u> ±0.091	0.669±0.089
SegMamba*	0.784±0.079 #	0.759±0.069	0.704±0.075	<u>0.859</u> ±0.067
VT-UNet*	0.785±0.094	0.759±0.081	0.709±0.087	0.845±0.079
TransBTS*	0.853 ±0.054	0.863 ±0.049	0.928 ±0.047	0.821±0.050
MCADS	0.821±0.057 #	0.819±0.051	0.843±0.051	0.798±0.050
GF-Net	0.793±0.041 #	0.798±0.045	0.781±0.044	0.814±0.039
MAD-Former	0.835±0.043 #	0.827±0.038	0.840±0.042	0.813±0.037
AAGN	0.801±0.039 #	0.795±0.041	0.771±0.033	0.820±0.043
dGLCN	0.812±0.051 #	0.807±0.049	0.819±0.044	0.792±0.046
IMDL(M)	0.802±0.047 #	0.781±0.045	0.708±0.038	0.871 ±0.047
NeurGNN (ours)	<u>0.849</u> ±0.041	<u>0.853</u> ±0.039	<u>0.878</u> ±0.043	0.837±0.044
NeurGNN (CI)	[0.7982, 0.9010]	[0.8047, 0.9029]	[0.825, 0.931]	[0.782, 0.892]

S2: Theoretical and Quantitative Evaluation of BRGC

We added quantitative analyses of intra class dispersion S_W , inter class center distance S_B , and their ratio R_{geo} .

$$S_W = \frac{1}{B} \sum_i \|z_i - c_{y_i}\|_2^2 \quad (1)$$

$$S_B = \frac{2}{K(K-1)} \sum_{k < l} \|c_k - c_l\|_2^2 \quad (2)$$

$$R_{\text{geo}} = \frac{S_B}{S_W + \epsilon} \quad (3)$$

As shown in Table 3, incorporating BRGC reduced S_W from 0.427 to 0.394, indicating more compact intra class distributions. S_B increased from 0.819 to 0.837, indicating greater separation between class centers. The increase in R_{geo} further demonstrates improved overall geometric discriminability.

Table 3: Quantitative distance analysis results.

Configuration	S_W	S_B	R_{geo}
None	0.427	0.819	1.918
BRGC	0.394	0.837	2.124

S3: Theoretical and Quantitative Evaluation of BRGC

we added a model efficiency analysis that systematically reports computational complexity, FLOPs, total parameter count, overall training time, average inference time per sample, and GPU memory consumption for different methods. As shown in Figure 4, all experiments were conducted under an identical hardware configuration using a single NVIDIA RTX 3090 GPU with 24 GB of memory. The batch size was fixed at 8 for all models to ensure a fair comparison, and both inference time and GPU memory consumption were measured under the same testing conditions. We selected several representative deep learning models for comparison, including CNN

based, Transformer based, Mamba based, and graph neural network based approaches. SegMamba exhibited relatively high parameter counts and GPU memory consumption, mainly because it employs a complete 3D encoder comprising multistage high dimensional feature extraction modules and Mamba sequence modeling units. Although NeurGNN incorporates brain region graph construction and graph reasoning, it performs computation using region level structural representations, thereby reducing GPU memory requirements while maintaining competitive predictive performance and computational efficiency.

Table 4: Model efficiency analysis.

Model	Total Training Time (s)	Average Inference Time per Sample (ms)	GPU Memory (MB)	Total Parameters (M)	FLOPs (G)	Computational complexity
3D_ResNet	5332.05	10.19	6730	63.511	83.541	$O(N)$
PCC_GAN	3656.95	3.86	5252	11.732	5.894	$O(N)$
SegMamba	22935.17	55.78	13622	80.184	87.146	$O(N)$
TransBTS	2939.47	3.21	7416	61.261	18.553	$O(N^2)$
VT_UNet	13294.40	59.98	22870	21.186	40.553	$O(N^2)$
dGLCN	10500.28	22.54	3571	10.241	46.47	$O(N)$
NeurGNN	1517.36	40.17	2530	80.696	44.098	$O(N)$